ANITA MADANI

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EXPERIENCE

Machine Learning Research Intern (MLR)

Apple – Video Computer Vision (VCV) Team

May 2026 – Sep 2026 Sunnyvale, CA

- Proposed and developed a **novel GRPO variant** for spatial and temporal grounding in VLMs, integrating SFT and pseudo-labeling into the post-training framework.
- Designed a **new persistent scene-graph memory architecture** for long-horizon egocentric video understanding, improving **VideoQA by 17.8 points** and achieving **8× longer average tracks** than evaluated SOTA trackers.
- Reduced input-token usage by **7.4**× versus caption-based baselines and **68.6**× versus full-frame baselines; optimized an **8B VLM** for on-device inference through quantization.

PhD Research Intern

NEC Laboratories America INC - Media Analytics Lab

Feb 2026 - May 2026

- Designed **generative agent-based AI models** using **diffusion frameworks** to create high-fidelity, reactive simulations for realistic scene synthesis and trajectory forecasting.
- Developed pedestrian simulation using **SMPL-X** and **3D Gaussian Splatting**, combining physics-based articulation, diffusion-based trajectory generation, and **video diffusion models** for realistic scene rendering.
- Achieved **70.3% behavior alignment** (vs. **14.9%** baseline) through VLM-based action verification and refinement.

Graduate Research Assistant

Johns Hopkins University, AIEM & VIU

2024 – Present Baltimore, MD

- Developed **cross-view registration and localization** methods for **3D reconstruction**, addressing large viewpoint shifts and **doppelgänger ambiguities** across ground, UAV, and satellite imagery.
- Designed **diffusion-based alignment and change-detection models** to distinguish scene changes from geometric misalignment and seasonal variation.
- Built **VLM-based scene-understanding models** and automated evaluation workflows using **Airflow, AWS, and Docker** for the multi-institutional **WRIVA** project.

Research Assistant

Sharif University of Technology

2021- 2022 Tehran, Iran

- Developed a **large-scale Persian semantic search engine** leveraging a **knowledge graph of 2M+ documents** and **Graph Neural Networks (GCN, GAT)** for semantic representation learning.
- Enhanced **retrieval quality and efficiency** through spectral clustering and PageRank, improving **Mean Reciprocal Rank (MRR) by 17%** and reducing **query latency by 23%**.

HONORS & AWARDS

- **Graduate Fellowship** — Whiting School of Engineering, Johns Hopkins University
- **Qualified for national Math and Informatics Olympiad** (2016)
- **Valedictorian** — Farzanegan 1 School (NODET, Middle & High School)

EDUCATION

Ph.D., Electrical & Computer Engineering

Johns Hopkins University

2023 – 03/2027 (expected) United States **GPA:** 4.00

MSE, Electrical & Computer Engineering

Johns Hopkins University

2023 - 2025 United States **GPA:** 3.92

B.Sc., Electrical Engineering

Sharif University of Technology

2019 – 2023 Iran **GPA:** 3.74

PUBLICATIONS

- **Madani et al.** SceneGraphMem: Learning Persistent Scene Graph Memory for Long-Horizon Video Understanding, *Preprint*, **CVPR 2027**.
- **Madani et al.** PedEditor: Pedestrian Trajectory and Behavior Editing via Generative Models, *Preprint*, **CVPR 2027**.
- **Gerami et al.**IceBerg Attention: Sublinear Long-Context Decoding With Mixture-Of-Gaussians, *Under Review*, **ICLR 2027**.
- **Madani et al.**DiffRegCD: Integrated Scene Alignment and Change Detection Using Diffusion Features. **WACV 2026**.
- **Madani et al.**Morphing Through Time: Diffusion-Based Bridging of Temporal Gaps for Robust Alignment in Change Detection. **WACV 2026**.
- **Gerami et al.***On The Application of Linear Attention in Multimodal Transformers*. **CVPRW 2026**.

SKILLS

Expertise

Computer Vision | Video Understanding | Vision-Language Models | Multimodal LLMs | VLM Post-Training | LLM Agents | Reinforcement Learning | Graph Learning | 3D Reconstruction | Diffusion/Generative Models | Visual Localization | Change Detection | Remote Sensing | Algorithms & DS | Optimization | Linear Algebra | Probability | Signal & Image Processing

Coding

Python (50k+ lines) | C (10k) | C++ (10k) | Bash (5k) | Java (5k) | MATLAB (3k) | Verilog (1k)

Tools

PyTorch | TensorFlow | Keras | scikit-learn | Hugging Face | ONNX | OpenCV | COLMAP | torchvision | TensorBoard | MLflow | Git | Docker | SLURM | Conda/Mamba | CMake | Linux | AWS | SQL